ONE8 FOOTWEAR E-COMMERCE DATABASE

Product Management – SQL Queries and Business Reports

1. Introduction

This document adapts the supplied Product Management sample for the ONE8 Footwear E-Commerce Database. It covers product queries, filtering, sorting, price searches, customer information, and business reports.

2. Database Tables

Table	Purpose
Category	Stores footwear category names and descriptions.
Products	Stores product name, brand, category, price, size, color, and stock.
Customers	Stores basic customer information.
Orders	Stores customer, order date, total amount, and order status.
Order_Details	Stores products and quantities included in each order.

3. SQL Queries Included

3.1 Basic SQL Queries

Display all products

```
SELECT * FROM Products;
```

Display selected product details

```
SELECT Product_ID, Product_Name, Price, Stock FROM Products;
```

Filter by category

```
SELECT Product_ID, Product_Name FROM Products WHERE Category_ID = 1;
```

Sort by price descending

```
SELECT Product_ID, Product_Name, Price FROM Products ORDER BY Price DESC;
```

Display unique colors

```
SELECT DISTINCT Color FROM Products;
```

3.2 Search Products Based on Price

Products priced between 2000 and 3500

```
SELECT Product_ID, Product_Name, Price FROM Products WHERE Price BETWEEN 2000 AND 3500 ORDER BY Price;
```

3.3 Retrieve Customer Information

Display all customers

```
SELECT * FROM Customers;
```

3.4 Apply Filtering Conditions

AND condition

```
SELECT Product_ID, Product_Name, Price, Stock FROM Products WHERE Price < 3500 AND Stock > 15;
```

OR condition

```
SELECT Product_ID, Product_Name, Category_ID FROM Products WHERE Category_ID = 1 OR Category_ID = 2;
```

NOT condition

```
SELECT Product_ID, Product_Name, Color FROM Products WHERE NOT Color = 'Black';
```

3.5 Basic Business Reports

Product price report

```
SELECT Product_ID, Product_Name, Price FROM Products ORDER BY Price DESC;
```

Customer order report

```
SELECT o.Order_ID, c.Customer_Name, o.Order_Date, o.Total_Amount, o.Order_Status FROM Orders o JOIN Customers c ON c.Custo
```

Product order report

```
SELECT p.Product_ID, p.Product_Name, SUM(od.Quantity) AS Total_Quantity_Ordered, SUM(od.Quantity * od.Price) AS Total_Sale
```

Customer-wise total orders

```
SELECT c.Customer_ID, c.Customer_Name, COUNT(o.Order_ID) AS Total_Orders FROM Customers c LEFT JOIN Orders o ON o.Customer
```

Customer-wise total purchase amount

```
SELECT c.Customer_ID, c.Customer_Name, COALESCE(SUM(o.Total_Amount), 0) AS Total_Purchase_Amount FROM Customers c LEFT JOI
```

4. SQL Concepts Used

CREATE DATABASE, CREATE TABLE, INSERT, SELECT, WHERE, BETWEEN, AND, OR, NOT, ORDER BY, DISTINCT, JOIN, LEFT JOIN, SUM(), COUNT(), GROUP BY, COALESCE(), PRIMARY KEY, and FOREIGN KEY.

5. How to Run

Open ONE8_Product_Management.sql in MySQL Workbench, connect to your MySQL server, and click the lightning-bolt Execute button. The script creates the database and sample tables, inserts sample records, and runs all queries. It drops and recreates these five sample tables, so do not use it where existing data in those tables must be preserved.

6. Conclusion

These queries help the ONE8 database display and search footwear products, retrieve customer information, filter and sort records, and generate order and purchase reports.